

KeraNova Statistical Analysis Plan

Primary target: observed 12-month CDVA in logMAR
KeraNova Research Team | Version 1.0 | 1 October 2026

1 Objectives, estimands and populations

The primary objective is prediction of observed 12-month CDVA (logMAR) from baseline measurements in eligible treated eyes. Primary performance is out-of-sample mean absolute error (MAE). The secondary cohort comparison estimates an observed baseline-adjusted difference in 12-month CDVA between TG-PRK + ACXL and ACXL alone. It is an association, not a randomized or causal treatment effect.

Screening population: all extracted records with individual exclusion/pending reasons. Descriptive population: all verified eligible eyes, including those without the primary outcome. Outcome population: eligible eyes with a dated, verified CDVA endpoint in the 10–14-calendar-month window. Main model population: the outcome population with a verified patient ID, baseline CDVA, age, recorded sex, cohort and baseline Kmax/TCT. Each analysis records eyes and unique patients separately. Both eligible eyes remain linked to the same patient in all splits and resampling.

Endpoint	Definition and denominator
Primary	CDVA at the valid visit nearest the 12-month anniversary within 10–14 calendar months; tie selects earlier visit. Denominator: eyes with this observed outcome.
Change	Follow-up minus baseline for the same eye. Negative CDVA/UDVA logMAR change indicates improvement; denominators require both observed values.
Secondary	UDVA, refraction, K1/K2/Kmax/TCT and documented complications, each using its available-case denominator.
Safety	Number of eyes with a documented event / number with documented assessment; separately show unknown/not assessed. Do not treat blanks as no event.

2 Before analysis and feasibility rules

First verify the AI-extension governance determination, index/visit dates, eligibility, procedure definitions, patient linkage, visual-acuity units and data dictionary. The preliminary CSV contains only two nominal 12-month corrected-vision observations and lacks patient codes/dates; it cannot support the prespecified model or a credible primary treatment comparison. On that extract, execute quality/descriptive reporting only and label model outputs not run.

There is no universal minimum-row rule that certifies adequacy. Before development, the biostatistician documents independent-patient count, complete endpoint count, six candidate slopes, outcome spread, cohort/predictor variation, missingness and an anticipated-performance/precision-based sample-size assessment [2]. Any sensitivity model with additional missingness indicators requires a separate complexity assessment. If grouped validation or calibration estimation is unstable, report that limitation or omit the unsupported estimate with a reason; do not present a successful numerical fit as proof of feasibility.

3 Descriptive analysis and missing data

Report continuous baseline/outcome values with available n, mean and SD, and median and interquartile range; report categories as n/N (%) and missing n. Tabulate source availability and adjudication exclusions before producing

clinical results. Describe baseline differences using standardized differences rather than significance testing to decide which variables enter the model.

The main model is complete-case for all six verified predictors and observed endpoint. Report exclusions and compare included versus excluded eyes descriptively. Do not impute CDVA outcomes, use last observation carried forward, substitute UDVA, or convert low-vision categories into assumed numeric targets.

If authorized data contains sufficient observed predictors/outcomes and complete-case loss is material, run a separately labelled sensitivity pipeline: require verified patient ID, cohort, baseline CDVA, age and sex; impute missing baseline Kmax/TCT using training-partition medians and add a missingness indicator for each. Learn imputation and scaling separately inside every inner and outer training partition. A predictor entirely unavailable or without meaningful variation prevents this sensitivity analysis. This evaluates a specific pragmatic preprocessing approach; it does not solve informative missingness.

4 Prespecified prediction models

All models target continuous logMAR. No class balancing, oversampling or arbitrary dichotomization is used. Fit all three models on the same observations for paired performance comparisons. Model development uses eye-weighted loss; patients remain the independent resampling unit. A patient-weighted evaluation, assigning each patient's total weight equally across their included eyes, is a sensitivity analysis. No model choice is made from the outer test results.

Model	Specification
Main	Ridge linear regression: $CDVA_{12} \sim CDVA_0 + age + sex + treatment + Kmax_0 + TCT_0$. Intercept unpenalized; six slopes, with no interactions/nonlinear terms.
Clinical benchmark	Ordinary linear regression: $CDVA_{12} \sim CDVA_0 + treatment$. Same model-population records and validation partitions as main model.
Mean reference	Predict the training-partition mean CDVA12 for every test eye.
Coding	TG-PRK + ACXL=1, ACXL alone=0. Recorded male=1 and female=0 for the source categories; other/unknown categories trigger a documented coding amendment and complexity reassessment, rather than inference.
Preprocessing	Continuous predictors use training-partition mean and SD. No test-partition centering, imputation, outlier-based deletion or feature selection.
Penalty	For standardized design matrix Z , minimize mean squared residual + $\lambda \times \text{sum of squared slopes}$. Grid $\lambda = \{0.0001, 0.001, 0.01, 0.1, 1, 10, 100\}$. Pick lowest inner-CV MAE; ties choose larger λ . Store software parameter mapping to this objective.

5 Internal validation without information leakage

Use 20 repeats of five-fold outer cross-validation grouped by coded patient ID. Allocate patients to folds with treatment balance where possible while keeping every patient's eyes together. Within each outer training set, use four-fold grouped inner cross-validation to choose the ridge penalty. Fit preprocessing and the selected model on the outer training set only, then predict every eye in its held-out patients. Save the fold allocation and each prediction. Use a recorded base random seed of 20261001 with deterministic derived repeat/resample seeds.

Require each training fold to contain the relevant treatment/sex levels, nonzero outcome variation and usable continuous predictors. Do not force an invalid five/four-fold analysis when the patient sample cannot support it. Any simpler resampling design requires a dated statistical amendment before performance is examined. A single random holdout is not the planned internal validation method.

For each repeat, pool its outer held-out predictions, so each eye contributes once to the repeat's MAE/RMSE and calibration. The main estimates are the mean of the 20 repeat-level metrics. Do not calculate confidence intervals from the distribution of fold scores or treat 20 predictions for one eye as 20 observations. A display may use each eye's mean held-out prediction, clearly identified as a visualization; metrics remain defined at repeat level.

After evaluation is complete, select the penalty with grouped inner validation on the entire development sample and refit the locked pipeline on those data for research demonstration. The final fit does not create a new validation estimate. Publish the intercept, coefficients, coding, scaling constants, penalty/objective and preprocessing necessary to reproduce predictions.

6 Performance, uncertainty and explanations

For approximate 95% CIs, perform 1,000 patient-cluster bootstrap resamples. Retain all eligible eyes of each sampled patient; all duplicate copies of a source patient remain in the same grouped fold. Repeat the entire preprocessing/tuning/nested evaluation pipeline and paired comparison within each resample, then use the 2.5th and 97.5th percentiles of the resample metrics. Record failed/infeasible resamples and reasons; report successful count and uncertainty limitations. If this procedure cannot yield stable feasible resamples, label intervals not estimable and do not replace them with naive fold-based intervals.

These bootstrap CIs are exploratory approximations: patient count can be small, resamples can lack strata/variation, and repeated validation has dependence. Wide or unstable intervals must remain visible. No significance threshold or desired accuracy number is used as a hackathon pass condition. Evidence of usefulness requires error improvement, reasonable calibration, stability and independent future validation.

Measure	Reporting rule
MAE / RMSE	MAE = mean absolute observed-minus-predicted logMAR; RMSE = square root of mean squared residual. Both are reported in logMAR, with lower values indicating smaller error.
Calibration	Regress observed outcome on held-out prediction: $y = a + b \times \text{prediction}$. Report intercept a (ideal 0), slope b (ideal 1), and mean prediction error (prediction minus observed; ideal 0). Omit slope if predictions have insufficient variation.
Incremental performance	Report main-minus-benchmark MAE and RMSE using the same resamples/eyes. Negative differences favour main model prediction error; no clinical superiority claim follows.
Prediction uncertainty	Do not confuse a CI for average performance or the predicted mean with a 95% interval for an individual outcome. Without a separately validated interval method, do not display individual 95% prediction intervals.
Explanation	Report standardized and original-unit ridge coefficients with resampling stability. Display associations, not causal effects. Coefficient rank alone is not a definitive importance or biological mechanism.
Subgroup checks	Describe errors by treatment and sex only with eyes/patients and uncertainty shown. Small cells are labelled too sparse for inference, not proof of fairness.

7 Secondary cohort comparison and safety

Describe baseline and follow-up measurements by verified treatment cohort. For the principal secondary comparison fit $CDVA_{12} \sim \text{treatment} + CDVA_0$, using eyes with both observed values and verified patient linkage. The treatment coefficient represents the adjusted observed mean logMAR difference (combined minus ACXL alone). Obtain an approximate 95% CI by 1,000 patient-cluster bootstrap refits. If the sample cannot support the regression or clustered resampling, report unadjusted descriptive summaries with a clear reason. This comparison is not the main prediction model and does not tune its features.

For each remaining continuous outcome report paired change by cohort and an unadjusted difference in mean change with a patient-cluster bootstrap CI where feasible. Do not use an unpaired test on paired baseline/follow-up values or independent-eye standard errors for fellow eyes. These multiple secondary summaries are exploratory; emphasize effect sizes and CIs without confirmatory p-value claims.

Complications are descriptive n/N (%) with unknown assessment shown. Show documented CDVA worsening and improvement as continuous change; a categorical line-loss threshold requires an agreed chart/measurement definition before analysis. Haze grades are reported as recorded after harmonization. No haze-resolution survival curve is produced without verified dates and censoring rules.

8 Sensitivity analyses and deviations

Every deviation states its date, rationale, approver and whether outcomes/performance were known. New image models, outcomes, feature sets or algorithms belong to separately labelled exploratory work. Public diagnostic datasets cannot be presented as external validation of 12-month post-treatment CDVA prediction.

- One eye per patient: select the first eligible index procedure, with a same-date bilateral tie resolved OD before OS; select without observing outcomes. Repeat main evaluation only if still feasible.
- Missing Kmax/TCT: run the prespecified training-only imputation sensitivity in section 3 and compare its included population with complete cases.
- Visual-acuity comparability: repeat summaries for observations with documented comparable correction/testing conditions; report exclusions.
- Intervening procedures: retain real-world outcomes in the main descriptive population and show a sensitivity excluding documented additional corneal procedures before the endpoint.
- Baseline timing: show the surgery-to-baseline interval distribution; any exclusion window for distant baseline must be clinically endorsed and recorded before performance is inspected.
- Patient weighting: repeat prediction metrics with equal total patient weight to examine the influence of bilateral-eye inclusion.

9 Separate public CXL-only feasibility analysis

If used, register the public supporting workbook of Wajnsztajn et al. [5] as a distinct source and experiment before fitting. Target the source-labelled one-year CDVA logMAR, not final-visit logMAR and not a verified 10–14-month KKESH endpoint. Candidate inputs are source baseline CDVA, age, recorded sex, Kmax and source-reported central pachymetry; do not rename pachymetry as TCT. Use a five-slope ridge model, baseline-CDVA-only clinical benchmark and training-mean reference. Source CXL protocol mix is summarized and accelerated-only results may be a prespecified sensitivity if adequate.

Apply adult/quality screening, source-ID grouping, complete-case predictors/outcome, training-only standardization and the same nested tuning/performance definitions as sections 4–6. Audit the patient ID with three records and negative follow-up metadata; exclude unresolved identity/chronology records conservatively with reasons while retaining their raw values. Because actual visit dates and laterality are absent, endpoint-window and eye-identity limitations remain explicit. Save its own population, dictionary and results under public feasibility; no merging with KKESH data, no TG-PRK comparison, no external-validation label. No public-model fitting or performance is claimed by this protocol.

10 Reproducibility and reporting

SPSS 26 may supply descriptive/cohort outputs where available; an audited R or Python pipeline supplies grouped modelling and nested resampling. Save actual software/package versions, source file hash, code version, seeds, fold assignments, correction log, outcome selection log, model artefact and per-eye held-out predictions within authorized storage. Direct identifiers are excluded from reports, code and figures.

All table values reconcile to a saved analysis population and denominator. Check visual-acuity and change calculations independently, patient-group separation in every split, training-only preprocessing, and raw-versus-transformed traceability. Review clinical plausibility and the TRIPOD+AI reporting checklist; use PROBAST+AI to document development/evaluation concerns. Unexecuted outputs are labelled not run, not zero.

11 References

- [1] Collins GS et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. BMJ. 2024;385:e078378. <https://doi.org/10.1136/bmj-2023-078378>
- [2] Riley RD et al. Calculating the sample size required for developing a clinical prediction model. BMJ. 2020;368:m441. <https://doi.org/10.1136/bmj.m441>
- [3] Moons KGM et al. PROBAST+AI: updated quality, risk of bias, and applicability assessment. BMJ. 2025;388:e082505. <https://doi.org/10.1136/bmj-2024-082505>
- [4] Vabalas A et al. Machine learning algorithm validation with a limited sample size. PLoS One. 2019;14:e0224365. <https://doi.org/10.1371/journal.pone.0224365>
- [5] Wajnsztajn D et al. Predicting factors for the efficacy of cross-linking for keratoconus. PLoS One. 2022;17:e0263528. <https://doi.org/10.1371/journal.pone.0263528>

